

Physics of Complex Systems: Exercise

Leopold Schaller and Daniel Lender

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Task 4.1: Convolution product

(a)

The probability density $p(x)$ is given and its characteristic function will be calculated via

$$\varphi(t) = E[e^{itX}] = \int_{-\infty}^{\infty} e^{itx} p(x) dx.$$

As $p(x) = 0$ is valid for $x < 0$, we will integrate from 0 to ∞ .

$$\varphi(t) = \int_0^{\infty} e^{itx} w e^{-wx} dx = w \int_0^{\infty} e^{-(w-it)x} dx.$$

For $\text{Re}(w - it) = w > 0$ and with $a = w - it$ we obtain

$$\int_0^{\infty} e^{-ax} dx = \frac{1}{a} = \frac{1}{w - it}.$$

Finally, we obtain the characteristic function

$$\boxed{\varphi(t) = \frac{w}{w - it}}.$$

(b) and (c)

The two fold convolution is defined as

$$p_{(x)}^{*2} = (p * p)(x) = \int_0^x p(y) p(x - y) dy.$$

Higher orders are calculated by the same formula, using

$$(p * p * p)(x) = p_{(x)}^{*3} = (p * p_{(x)}^{*2})(x) \text{ and } p_{(x)}^{*N} = (p * p_{(x)}^{*N-1})(x)$$

We calculate $p_{(x)}^{*2}$ explicitly and get

$$(p * p)(x) = \int_0^x w e^{-wy} w e^{-w(x-y)} dy = w^2 e^{-wx} \int_0^x dy = w^2 x e^{-wx}$$

For the three fold convolution product, we get

$$p_{(x)}^{*3} = \int_0^x w^2 y e^{-wy} \cdot w e^{-w(x-y)} dy = w^3 e^{-wx} \int_0^x y dy = \frac{w^3 x^2}{2} e^{-wx}.$$

Comparing the two solutions

$$p^{*1}(x) = w e^{-wx}, \quad p^{*2}(x) = w^2 x e^{-wx}, \quad p^{*3}(x) = \frac{w^3 x^2}{2} e^{-wx}.$$

we might observe a systematic behaviour according to

$$p^{*N}(x) = \frac{w^N x^{N-1}}{(N-1)!} e^{-wx}.$$

because the exponential e^y part will cancel out every time and what remains is an integral over a polynomial, which increases its power by one in every convolution step. To show this guess, we do a prove via induction:

Assume that for some N the formula holds:

$$p^{*N}(x) = \frac{w^N x^{N-1}}{(N-1)!} e^{-wx}.$$

We now compute the $(N+1)$ -fold convolution:

$$p^{*(N+1)}(x) = \int_0^x p^{*N}(y) p(x-y) dy.$$

Insert the induction hypothesis and $p(x) = we^{-wx}$:

$$= \int_0^x \frac{w^N y^{N-1}}{(N-1)!} e^{-wy} \cdot we^{-w(x-y)} dy = \frac{w^{N+1}}{(N-1)!} e^{-wx} \int_0^x y^{N-1} dy.$$

The remaining integral evaluates to

$$\int_0^x y^{N-1} dy = \frac{x^N}{N}.$$

Hence,

$$p^{*(N+1)}(x) = \frac{w^{N+1}}{(N-1)!} \frac{x^N}{N} e^{-wx} = \frac{w^{N+1} x^N}{N!} e^{-wx}.$$

$$p^{*(N+1)}(x) = \frac{w^{N+1} x^N}{N!} e^{-wx}.$$

This proves the formula by induction.

(d)

For a one fold probability distribution $p(x) = we^{-wx}$ we can calculate the expectation value and the variance to

$$\mu_1 = \frac{1}{w} \text{ and } \sigma_1^2 = \frac{1}{w^2}$$

As the experiment is statistically independent, we will obtain for $p^{*(N)}(x)$

$$\mu_N = \frac{N}{w} \text{ and } \sigma_N^2 = \frac{N}{w^2}$$

With σ_N and μ_N , we can build a normal distribution and compare it to the probability density $p^{*(N)}(x)$, which is shown in Fig. 1 for $N = 100$ and 50 . Both curves match very good to each other, showing the Central Limit Theorem works.

Task 4.2: Harmonic oscillator

(a)

The Liouville Operator L_{nm} is defined via

$$\frac{d}{dt} P_n(t) = \sum_m L_{nm} P_m(t),$$

where we can insert the rates given in the sheet. We assume only transitions from n to $n \pm 1$ are possible and obtain for $n > 0$:

$$\frac{d}{dt} P_n(t) = w_{\uparrow} P_{n-1}(t) + w_{\downarrow} P_{n+1}(t) - (w_{\uparrow} + w_{\downarrow}) P_n(t),$$

As no eigenstate $|-1\rangle$ exists, the first energy level $n = 0$ is a special case:

$$\frac{d}{dt} P_0(t) = w_{\downarrow} P_1(t) - w_{\uparrow} P_0(t)$$

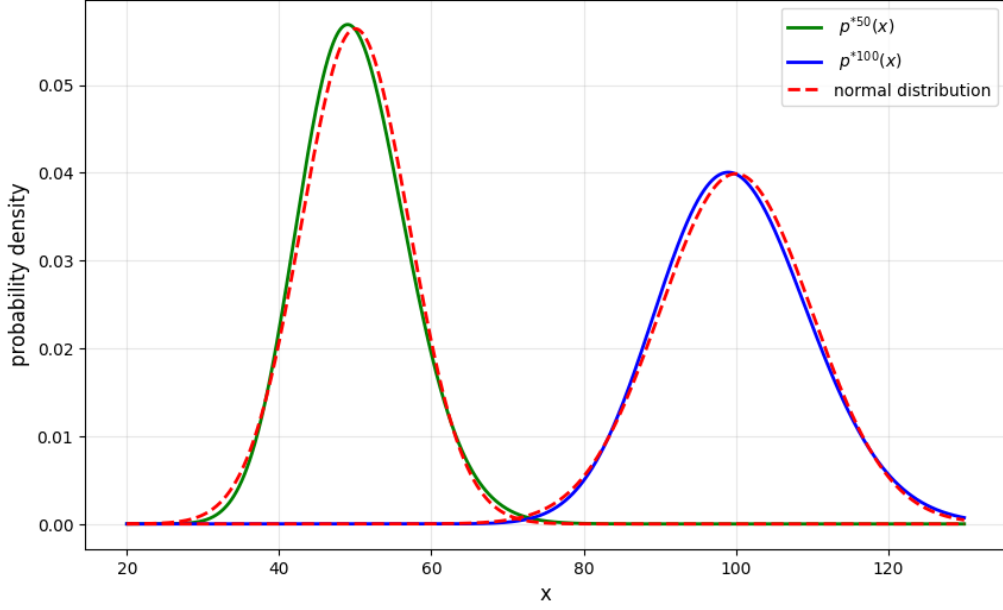


Figure 1: Calculated probability distribution and corresponding gaussian distribution

The entire matrix reads the following:

$$\begin{pmatrix} -w_{\uparrow} & w_{\downarrow} & 0 & 0 & \dots \\ w_{\uparrow} & -(w_{\uparrow} + w_{\downarrow}) & w_{\downarrow} & 0 & \dots \\ 0 & w_{\uparrow} & -(w_{\uparrow} + w_{\downarrow}) & w_{\downarrow} & \dots \\ 0 & 0 & w_{\uparrow} & -(w_{\uparrow} + w_{\downarrow}) & \dots \\ 0 & 0 & 0 & w_{\uparrow} & \dots \\ \dots & \dots & \dots & \dots & \dots \end{pmatrix}$$

(b)

The left eigenvector is $\sum = (1, 1, 1, \dots)$ and ensures that every column has a vanishing sum. The right eigenvector is calculated for the equations

$$\begin{aligned} n=0 : 0 &= w_{\downarrow}P_1(t) - w_{\uparrow}P_0(t) \Rightarrow P_1(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right) P_0(t) \\ n=1 : 0 &= w_{\uparrow}P_0(t) + w_{\downarrow}P_2(t) - (w_{\uparrow} + w_{\downarrow})P_1(t) = w_{\downarrow}P_1(t) + w_{\downarrow}P_2(t) - (w_{\uparrow} + w_{\downarrow})P_1(t) \\ &= w_{\downarrow}P_2(t) - w_{\uparrow}P_1(t) \Rightarrow P_2(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right) P_1(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right)^2 P_0(t) \end{aligned}$$

now we assume for a state $n > 1$:

$$P_{n+1}(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right) P_n(t) \Rightarrow P_{n+1}(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right)^{n+1} P_0(t)$$

insert the induction hypothesis into the equation for the state $n + 1$:

$$\begin{aligned} 0 &= w_{\uparrow}P_n(t) + w_{\downarrow}P_{n+2}(t) - (w_{\uparrow} + w_{\downarrow})P_{n+1}(t) = w_{\downarrow}P_{n+1}(t) + w_{\downarrow}P_{n+2}(t) - (w_{\uparrow} + w_{\downarrow})P_{n+1}(t) \\ &= w_{\downarrow}P_{n+2}(t) - w_{\uparrow}P_{n+1}(t) \Rightarrow P_{n+2}(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right) P_{n+1}(t) = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right)^2 P_n(t) \\ \Rightarrow P_{n+2} &= \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right)^{n+2} P_0(t) \end{aligned}$$

Now we proved

$$P_n = \left(\frac{\omega_{\uparrow}}{\omega_{\downarrow}}\right)^n P_0$$

by induction. Where P_n is the n -th component of the right eigenvector to the eigenvalue zero.

(c)

The normalization condition yields :

$$1 = \sum \cdot \vec{P}_n = \sum_{n=0}^{\infty} P_n = P_0 \sum_{n=0}^{\infty} \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)^n = P_0 \left(\frac{1}{1 - \frac{w_{\uparrow}}{w_{\downarrow}}} \right)$$
$$\Rightarrow P_0 = 1 - \frac{w_{\uparrow}}{w_{\downarrow}}$$

The calculation contains a geometric series, it only converges, if the condition $w_{\uparrow} < w_{\downarrow}$ is fulfilled. Now we can express the components of the right eigenvector to the eigenvalue zero only dependent from $w_{\uparrow}$ and $w_{\downarrow}$:

$$P_n = \left(1 - \frac{w_{\uparrow}}{w_{\downarrow}} \right) \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)^n = \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)^n - \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)^{n+1}$$

(d)

The probability distribution in a canonical ensemble is

$$p_n \propto \exp \left(\frac{E_n}{k_B T} \right) = \exp \left(\frac{\hbar \omega}{2 k_B T} \right) \cdot \exp \left(\frac{\hbar \omega n}{k_B T} \right)$$
$$= A \cdot \left[\exp \left(\frac{\hbar \omega}{k_B T} \right) \right]^n$$

Furthermore we know

$$p_n \propto \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)^n$$

Comparing the base of the exponent for both terms leads to this expression:

$$\frac{w_{\uparrow}}{w_{\downarrow}} = e^{\frac{\hbar \omega}{k_B T}} \quad (1)$$

$$\Rightarrow \ln \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right) = - \frac{\hbar \omega}{k_B T} \quad (2)$$

solving for T yields

$$T = - \frac{\hbar \omega}{k_B \ln \left(\frac{w_{\uparrow}}{w_{\downarrow}} \right)} = \frac{\hbar \omega}{k_B \ln \left(\frac{w_{\downarrow}}{w_{\uparrow}} \right)} \quad (3)$$